

Лабораторная работа №5

Управление системными службами

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Цель работы

Получить навыки управления системными службами операционной системы посредством systemd.

Ход выполнения работы

Управление сервисом vsftpd

```
rkogengar@rishardkogengar:~$ su
Password:
root@rishardkogengar:/home/rkogengar# systemctl status vsftd
Unit vsftd.service could not be found.
root@rishardkogengar:/home/rkogengar# dnf -y install vsftpd
Rocky Linux 10 - BaseOS                               8.5 kB/s | 3.9 kB    00:00
Rocky Linux 10 - BaseOS                               28 MB/s | 18 MB     00:00
Rocky Linux 10 - AppStream                             14 kB/s | 3.9 kB    00:00
Rocky Linux 10 - AppStream                             3.5 MB/s | 2.1 MB    00:00
Rocky Linux 10 - Extras                               9.1 kB/s | 3.1 kB    00:00
Rocky Linux 10 - Extras                               16 kB/s | 4.9 kB    00:00
Dependencies resolved.

=====
Package           Architecture      Version           Repository        Size
=====
Installing:
vsftpd            x86_64            3.0.5-9.el10     appstream         170 k

Transaction Summary
=====
Install 1 Package

Total download size: 170 k
Installed size: 348 k
Downloading Packages:
vsftpd-3.0.5-9.el10.x86_64.rpm                        3.4 MB/s | 170 kB    00:00
-----
Total                                                    577 kB/s | 170 kB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing
```

```
root@rishardkogengar:/home/rkogengar# systemctl start vsftpd
root@rishardkogengar:/home/rkogengar# systemctl status vsftpd
● vsftpd.service - Vsftpd ftp daemon
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; disabled; preset: disabled)
   Active: active (running) since Sat 2025-09-20 12:27:14 MSK; 32s ago
 Invocation: c99d9b43b6e04b85a59db67cb9422f33
   Process: 4171 ExecStart=/usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf (code=exited, status=0/SUCCESS)
  Main PID: 4175 (vsftpd)
    Tasks: 1 (limit: 24779)
   Memory: 748K (peak: 1.2M)
      CPU: 3ms
   CGroup: /system.slice/vsftpd.service
           └─4175 /usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf

Sep 20 12:27:14 rishardkogengar.localdomain systemd[1]: Starting vsftpd.service - Vsftpd ftp daemon:
Sep 20 12:27:14 rishardkogengar.localdomain systemd[1]: Started vsftpd.service - Vsftpd ftp daemon:
lines 1-14/14 (END)
```

Рис. 2: Запуск службы vsftpd

```
root@rishardkogengar:/home/ikogengar# systemctl enable vsftpd
Created symlink '/etc/systemd/system/multi-user.target.wants/vsftpd.service' → '/usr/lib/systemd/system/vsftpd.service'.
root@rishardkogengar:/home/ikogengar# systemctl status vsftpd
● vsftpd.service - Vsftpd ftp daemon
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2025-09-20 12:27:14 MSK; 2min 6s ago
 Invocation: c99d9b43b6e04b85a59db67cb9422f33
    Main PID: 4175 (vsftpd)
      Tasks: 1 (limit: 24779)
     Memory: 748K (peak: 1.2M)
        CPU: 3ms
    CGroup: /system.slice/vsftpd.service
            └─4175 /usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf

Sep 20 12:27:14 rishardkogengar.localdomain systemd[1]: Starting vsftpd.service - Vsftpd ftp daemon>
Sep 20 12:27:14 rishardkogengar.localdomain systemd[1]: Started vsftpd.service - Vsftpd ftp daemon>
lines 1-13/13 (END)
```

Рис. 3: Добавление в автозапуск

```
root@rishardkogengar:/home/rkogengar# systemctl disable vsftpd
Removed '/etc/systemd/system/multi-user.target.wants/vsftpd.service'.
root@rishardkogengar:/home/rkogengar# systemctl status vsftpd
● vsftpd.service - Vsftpd ftp daemon
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; disabled; preset: disabled)
   Active: active (running) since Sat 2025-09-20 12:27:14 MSK; 2min 33s ago
 Invocation: c99d9b43b6e04b85a59db67cb9422f33
    Main PID: 4175 (vsftpd)
      Tasks: 1 (limit: 24779)
     Memory: 748K (peak: 1.2M)
        CPU: 3ms
      CGroup: /system.slice/vsftpd.service
              └─4175 /usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf

Sep 20 12:27:14 rishardkogengar.localdomain systemd[1]: Starting vsftpd.service - Vsftpd ftp daemo>
Sep 20 12:27:14 rishardkogengar.localdomain systemd[1]: Started vsftpd.service - Vsftpd ftp daemo>
lines 1-13/13 (END)
```

Рис. 4: Отключение автозапуска

Управление сервисом vsftpd

```
root@rishardkogengar:/home/rkogengar# ls /etc/systemd/system/multi-user.target.wants/
atd.service      cups.service      ModemManager.service  sssd.service
auditd.service   firewalld.service NetworkManager.service tuned.service
audit-rules.service irqbalance.service remote-cryptsetup.target vboxadd.service
avahi-daemon.service kdump.service      remote-fs.target       vboxadd-service.service
chronyd.service  libstoragemgmt.service rsyslog.service        vmttoolsd.service
crond.service    mcelog.service     smartd.service
cups.path        mdmonitor.service  sshd.service

root@rishardkogengar:/home/rkogengar# systemctl enable vsftpd
Created symlink '/etc/systemd/system/multi-user.target.wants/vsftpd.service' → '/usr/lib/systemd/system/vsftpd.service'.

root@rishardkogengar:/home/rkogengar# ls /etc/systemd/system/multi-user.target.wants/
atd.service      cups.service      ModemManager.service  sssd.service
auditd.service   firewalld.service NetworkManager.service tuned.service
audit-rules.service irqbalance.service remote-cryptsetup.target vboxadd.service
avahi-daemon.service kdump.service      remote-fs.target       vboxadd-service.service
chronyd.service  libstoragemgmt.service rsyslog.service        vmttoolsd.service
crond.service    mcelog.service     smartd.service         vsftpd.service
cups.path        mdmonitor.service  sshd.service

root@rishardkogengar:/home/rkogengar# systemctl status vsftpd
● vsftpd.service - Vsftpd ftp daemon
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2025-09-20 12:27:14 MSK; 4min 1s ago
 Invocation: c99d9b43b6e04b85a59db67cb9422f33
    Main PID: 4175 (vsftpd)
      Tasks: 1 (limit: 24779)
     Memory: 748K (peak: 1.2M)
        CPU: 3ms
    CGroup: /system.slice/vsftpd.service
            └─4175 /usr/sbin/vsftpd /etc/vsftpd/vsftpd.conf
```

```
Sep 20 12:27:14 rishardkogengar.localdomain systemd[1]: Starting vsftpd.service - Vsftpd ftp daemon>
Sep 20 12:27:14 rishardkogengar.localdomain systemd[1]: Started vsftpd.service - Vsftpd ftp daemon>
lines 1-13/13 (END)
```

```
● └─proc-sys-fs-binfmt_misc.automount
○ └─selinux-autorelabel-mark.service
● └─sys-fs-fuse-connections.mount
● └─sys-kernel-config.mount
● └─sys-kernel-debug.mount
● └─sys-kernel-tracing.mount
○ └─systemd-ask-password-console.path
○ └─systemd-binfmt.service
○ └─systemd-boot-random-seed.service
○ └─systemd-conext.service
○ └─systemd-firstboot.service
○ └─systemd-hibernate-clear.service
○ └─systemd-hwdb-update.service
○ └─systemd-journal-catalog-update.service
● └─systemd-journal-flush.service
● └─systemd-journald.service
○ └─systemd-machine-id-commit.service
root@rishardkogengar:/home/rkogengar# systemctl list-dependencies vsftpd --reverse
vsftpd.service
● └─multi-user.target
● └─graphical.target
root@rishardkogengar:/home/rkogengar#
```

Рис. 6: Обратные зависимости vsftpd

Конфликты юнитов: iptables и firewalld

```
-----gengar, home, gengar-----
root@rishardkogengar:/home/rkogengar# dnf -y install iptables\*
Last metadata expiration check: 0:06:33 ago on Sat 20 Sep 2025 12:26:41 PM MSK.
Package iptables-libs-1.8.11-8.el10_0.x86_64 is already installed.
Package iptables-nft-1.8.11-8.el10_0.x86_64 is already installed.
Dependencies resolved.
=====
Package                Architecture    Version           Repository        Size
=====
Installing:
iptables-devel          x86_64          1.8.11-8.el10_0   appstream         17 k
iptables-nft-services  noarch          1.8.11-8.el10_0   appstream         24 k
iptables-utils          x86_64          1.8.11-8.el10_0   appstream         42 k
=====

Transaction Summary
=====
Install 3 Packages

Total download size: 82 k
Installed size: 142 k
Downloading Packages:
(1/3): iptables-devel-1.8.11-8.el10_0.x86_64.rpm           297 kB/s | 17 kB    00:00
(2/3): iptables-nft-services-1.8.11-8.el10_0.noarch.rpm   366 kB/s | 24 kB    00:00
(3/3): iptables-utils-1.8.11-8.el10_0.x86_64.rpm          210 kB/s | 42 kB    00:00
-----
Total: 153 kB/s | 83 kB    00:00
```

Рис. 7: Установка iptables

Конфликты юнитов: iptables и firewalld

```
root@rishardkogengar:/home/rkogengar#
root@rishardkogengar:/home/rkogengar# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-09-20 12:20:44 MSK; 12min ago
 Invocation: 3a847d72226e46689ae09f55fc7af21c
    Docs: man:firewalld(1)
   Main PID: 946 (firewalld)
     Tasks: 2 (limit: 24779)
    Memory: 48.6M (peak: 50.6M)
       CPU: 256ms
    CGroup: /system.slice/firewalld.service
            └─946 /usr/bin/python3 -sP /usr/sbin/firewalld --nofork --nopid

Sep 20 12:20:44 rishardkogengar.localdomain systemd[1]: Starting firewalld.service - firewalld - >
Sep 20 12:20:44 rishardkogengar.localdomain systemd[1]: Started firewalld.service - firewalld - d>
root@rishardkogengar:/home/rkogengar# systemctl status iptables
○ iptables.service - IPv4 firewall with iptables
   Loaded: loaded (/usr/lib/systemd/system/iptables.service; disabled; preset: disabled)
   Active: inactive (dead)
root@rishardkogengar:/home/rkogengar#
```

Рис. 8: Проверка статуса firewalld и iptables

Конфликты юнитов: iptables и firewalld

```
root@rishardkogengar: /home/rkogengar# systemctl start firewalld
root@rishardkogengar: /home/rkogengar# systemctl start iptables
root@rishardkogengar: /home/rkogengar# cat /usr/lib/systemd/system/firewalld.service
[Unit]
Description=firewalld - dynamic firewall daemon
Before=network-pre.target
Wants=network-pre.target
After=dbus.service
After=polkit.service
Conflicts=iptables.service ip6tables.service ebtables.service ipset.service
Documentation=man:firewalld(1)

[Service]
EnvironmentFile=-/etc/sysconfig/firewalld
ExecStart=/usr/sbin/firewalld --nofork --nopid $FIREWALLD_ARGS
ExecReload=/bin/kill -HUP $MAINPID
# supress to log debug and error output also to /var/log/messages
StandardOutput=null
StandardError=null
Type=dbus
BusName=org.fedoraproject.FirewallD1
KillMode=mixed
DevicePolicy=closed
KeyringMode=private
LockPersonality=yes
MemoryDenyWriteExecute=yes
PrivateDevices=yes
ProtectClock=yes
```

Конфликты юнитов: iptables и firewalld

```
root@rishardkogengar: /home/rkogengar#  
root@rishardkogengar: /home/rkogengar# cat /usr/lib/systemd/system/iptables.service  
[Unit]  
Description=IPv4 firewall with iptables  
AssertPathExists=/etc/sysconfig/iptables  
Before=network-pre.target  
Wants=network-pre.target  
  
[Service]  
Type=oneshot  
RemainAfterExit=yes  
ExecStart=/usr/libexec/iptables/iptables.init start  
ExecReload=/usr/libexec/iptables/iptables.init reload  
ExecStop=/usr/libexec/iptables/iptables.init stop  
Environment=BOOTUP=serial  
Environment=CONSOLETYPE=serial  
  
[Install]  
WantedBy=multi-user.target  
root@rishardkogengar: /home/rkogengar#
```

Рис. 10: Файл юнита iptables

```
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar# systemctl stop iptables.service  
root@rishardkogengar:/home/rkogengar# systemctl start firewalld.service  
root@rishardkogengar:/home/rkogengar# systemctl mask iptables  
Created symlink '/etc/systemd/system/iptables.service' → '/dev/null'.  
root@rishardkogengar:/home/rkogengar# systemctl start iptables  
Failed to start iptables.service: Unit iptables.service is masked.  
root@rishardkogengar:/home/rkogengar# systemctl enable iptables  
Failed to enable unit: Unit /etc/systemd/system/iptables.service is masked  
root@rishardkogengar:/home/rkogengar#
```

Рис. 11: Ошибка запуска замаскированного сервиса

Изолируемые цели

```
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar# cd /usr/lib/systemd/system  
root@rishardkogengar:/usr/lib/systemd/system# grep Isolate *.target  
ctrl-alt-del.target:AllowIsolate=yes  
default.target:AllowIsolate=yes  
emergency.target:AllowIsolate=yes  
exit.target:AllowIsolate=yes  
graphical.target:AllowIsolate=yes  
halt.target:AllowIsolate=yes  
initrd-switch-root.target:AllowIsolate=yes  
initrd.target:AllowIsolate=yes  
kexec.target:AllowIsolate=yes  
multi-user.target:AllowIsolate=yes  
poweroff.target:AllowIsolate=yes  
reboot.target:AllowIsolate=yes  
rescue.target:AllowIsolate=yes  
runlevel0.target:AllowIsolate=yes  
runlevel1.target:AllowIsolate=yes  
runlevel2.target:AllowIsolate=yes  
runlevel3.target:AllowIsolate=yes  
runlevel4.target:AllowIsolate=yes  
runlevel5.target:AllowIsolate=yes  
runlevel6.target:AllowIsolate=yes  
soft-reboot.target:AllowIsolate=yes  
system-update.target:AllowIsolate=yes  
root@rishardkogengar:/usr/lib/systemd/system#
```

Рис. 12: Список изолируемых целей

```
You are in rescue mode. After logging in, type "journalctl -xb" to view
system logs, "systemctl reboot" to reboot, or "exit"
to continue bootup.
Give root password for maintenance
(or press Control-D to continue):
root@rishardkogengar:~# systemctl isolate reboot.target _
```

Рис. 13: Изоляция цели reboot.target

```
rkogengar@rishardkogengar:~$ su
Password:
su: Authentication failure
rkogengar@rishardkogengar:~$ su
Password:
root@rishardkogengar:/home/rkogengar# systemctl get-default
graphical.target
root@rishardkogengar:/home/rkogengar# systemctl set-default multi-user.target
Removed '/etc/systemd/system/default.target'.
Created symlink '/etc/systemd/system/default.target' → '/usr/lib/systemd/system/multi-user.target'.
root@rishardkogengar:/home/rkogengar#
```

Рис. 14: Проверка цели по умолчанию

```
Rocky Linux 10.0 (Red Quartz)
Kernel 6.12.0-55.12.1.el10_0.x86_64 on x86_64

Web console: https://rishardkogengar.localdomain:9090/ or https://10.0.2.15:9090/

rishardkogengar login: root
Password:
Last login: Sat Sep 20 12:40:47 on pts/0
root@rishardkogengar:~# systemctl set-default graphical.target
Removed '/etc/systemd/system/default.target'.
Created symlink '/etc/systemd/system/default.target' -> '/usr/lib/systemd/system/graphical.target'.
root@rishardkogengar:~# _
```

Рис. 15: Установка graphical.target как цели по умолчанию

Итоги работы

В ходе работы были изучены основы управления сервисами в системе systemd, их установка, запуск, автозагрузка и удаление. Рассмотрены конфликты между юнитами, способы их разрешения, изолируемые цели и настройка цели по умолчанию.